

Douglas Gordon Daly

Principal AI Architect / AI Platform Engineering Lead

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PROFESSIONAL SUMMARY

Senior technical leader with deep expertise architecting enterprise AI/ML platforms, governed GenAI and tool-calling systems, and operational intelligence solutions. Built governed analytics assistants, monitoring frameworks, and reusable data/AI platforms with strong emphasis on controls, observability, and auditable execution. Experienced across cloud-native AI infrastructure, large-scale data platforms, executive alignment, and technical mentorship. Known for translating ambiguous business needs into scalable, reliable analytical systems that improve decision-making and operational performance.

CORE EXPERTISE

- AI Platform Architecture - enterprise AI platforms, governed analytics assistants, operational intelligence systems, reusable AI/analytics frameworks
- GenAI & Tool-Calling Systems - LLM applications, structured outputs, tool orchestration, RAG/GraphRAG concepts, prompt engineering, evaluation, traceability
- Governance & Operational Controls - responsible AI adoption, guardrails, auditable execution, analyst-in-the-loop workflows, policy-aware system design
- Observability & Reliability - monitoring frameworks, incident detection, alerting, operational health, SLIs/SLOs, SEV analysis, on-call-friendly design
- Cloud & Data Platforms - Python, SQL, BigQuery, AWS SageMaker/S3/Terraform, GCP, Docker, data profiling, validation, cross-system integration
- Technical Leadership - architecture direction, executive alignment, technical roadmaps, stakeholder translation, mentoring, developer enablement

PROFESSIONAL EXPERIENCE

Principal Consultant | Daly Engineers LLC | Jan 2026 - Present

Consolidated Edison (via ABC Consulting)

- Architected and delivered a governed analytics assistant leveraging Gemini, BigQuery, and tool-calling architectures, embedding analyst rules, operational controls, and auditable execution for AI-enabled construction and financial analytics.
- Developed automated frameworks for profiling, type inference, duplicate detection, candidate key analysis, and cross-system join validation across 100+ source tables and millions of operational and financial records, enhancing data quality and interoperability.
- Defined a phased analytical maturity roadmap for anomaly detection, construction analytics, and AI-enabled decision support, guiding strategic adoption of advanced analytics capabilities.

NBCUniversal (via Apex Systems)

- Architected AWS-based AI and graph analytics pipelines leveraging SageMaker, Terraform, S3, and containerized workloads to enable scalable, cloud-native analytics solutions.
- Evaluated and integrated emerging graph analytics and graph machine learning technologies—including Neptune, GraphStorm, Bedrock, and LangGraph—to support dependency analysis and operational intelligence initiatives.

AI Instructor | SimpliLearn & Interview Kickstart | March 2025 - Present

- Mentored engineers and technical professionals on AI architecture, governance, implementation tradeoffs, and enterprise adoption strategies, fostering cross-functional enablement and technical leadership.
- Delivered advanced instruction on transformer architectures, LLMs, RAG, agentic and multimodal AI, diffusion models, and enterprise AI governance, enabling practical understanding of generative and agentic AI development workflows.
- Led multi-month AI and machine learning training programs for software engineers and data scientists through university- and enterprise-sponsored continuing education initiatives, driving continuous learning and talent development.

Data Scientist | Intuit (via The Cydio Group) | Apr 2025 - Jul 2025

- Consolidated fragmented reporting into unified Tableau dashboards and automated executive reporting workflows using Python and SQL, significantly reducing report preparation time and driving operational efficiency.
- Collaborated with finance, marketing, and product stakeholders to translate attribution and performance data into actionable business recommendations, strengthening cross-functional alignment.

Data Scientist | Meta | October 2021 - January 2025

- Designed and developed the ReMon framework, empowering engineering teams to implement comprehensive monitoring and alerting systems without deep internal tooling expertise; adopted across Meta FinTech and enabled automatic detection of over 15 SEVs in 2024.
- Built forecasting and performance-management frameworks that translated engineering initiatives into projected reliability outcomes, directly informing planning, goal setting, and leadership decision-making across business pillars.
- Delivered impact sizing, ROI, and risk exposure analyses to guide investment and prioritization decisions for platform reliability, privacy compliance, authentication, testing infrastructure, and data platform initiatives.

Data Scientist | Samsung Electronics (via Harvey Nash) | Dec 2020 - Oct 2021

- Developed and deployed gradient-boosting propensity models to optimize digital marketing campaigns, driving a 146% increase in conversions and a 229% rise in campaign-attributed revenue while automating audience segmentation and targeting workflows for improved efficiency and scalability.

Head of Data Science | Facteus | Oct 2017 - Nov 2020

- Directed data science initiatives leveraging a 30B+ transaction dataset to develop analytical products and market intelligence capabilities through large-scale feature engineering, merchant intelligence, and consumer-spending analytics.

- Evaluated prospective data partnerships by assessing signal quality, coverage, and business value, supporting strategic investment and acquisition decisions.

Expert SEO Solutions Architect | Nike (via Intersoft Inc.) | Nov 2016 - Sep 2017

- Architected a scalable SEO and information-architecture framework for Nike's product discovery, aligning customer navigation with search engine indexation to support a rapidly growing product catalog.

Senior Manager, Operations Analysis | Capital One | Jan 2012 - Dec 2016

- Conceived, architected, and led adoption of RITHM, a real-time operational intelligence platform aggregating monitoring signals by business capability and system ownership, reducing major incident detection times from hours to minutes.
- Secured executive sponsorship and partnered with CTO-level leadership to define enterprise monitoring strategy, expanding Splunk-based prototypes into a scalable Kafka and Elastic Stack solution aligned with open-source modernization initiatives.
- Developed predictive capacity-management models for customer-facing banking platforms, enabling infrastructure planning and risk management through demand forecasting based on business-process and temporal drivers.

Senior Principal Systems Engineer / Program Manager | Raytheon | Mar 2005 - Dec 2011

- Led systems engineering organizations of 40-50 engineers specializing in modeling, simulation, analysis, and advanced aerospace systems, providing technical leadership, career development, and proposal support for programs valued up to \$50M.
- Turned around an over-budget classified fixed-price technology program by establishing execution plans, aligning engineering teams, and proactively managing technical and program risks; delivered under budget and nearly doubled the original profit target on a historically challenging delivery effort.

SELECTED AI PROJECTS

GraphRAG Engine & LoRA Coverage Assistant | GitHub | 2025

- Built a GraphRAG system for multi-hop benefits reasoning with fully traceable retrieval paths across entities, clauses, and supporting evidence.
- Fine-tuned a LoRA adapter on a 7B open-source LLM to enforce a strict JSON schema (decision/reason/disclaimer) for reliable downstream automation.

Agent-to-Agent Protocol MVPs | GitHub | 2026

- Agent Olympics: built a small evaluation runner that benchmarks a generalist agent vs discovered specialists across three task categories with explicit scoring criteria and summary reporting.
- AI PEMDAS: built an instrumented coordination workflow where an evaluator routes sub-steps to ADD vs MULT agents and logs calls for end-to-end traceability (FastAPI JSON-RPC).

AI-Assisted Resume Generation System | GitHub | 2026

- Built an evidence-driven resume generation pipeline that converts a master resume into canonical JSON, extracts candidate evidence, maps evidence to market-derived role signals, and assembles targeted resume variants through structured LLM workflows.
- Implemented versioned artifacts, human review checkpoints, traceable evidence IDs, and DOCX/PDF rendering controls to separate source evidence, content selection, positioning, and presentation layout.

EDUCATION

MBA | (UCLA Anderson School of Management): Top Honors | GPA: 3.9

M.S., Applied Statistics | (Rochester Institute of Technology): GPA: 3.9

M.S., Electrical Engineering | (University of Washington): GPA: 3.7

B.S., Physics (Minor: Mathematics) | (Humboldt State University): GPA: 3.4

PATENTS & RECOGNITION

- Lead Inventor, U.S. Patent #8635622: Method and System for Resource Management Using Fuzzy Logic Timeline Filling (2014)
- Capital One Business Technology Innovation Award (2014)
- Conference Chair & Speaker - Customer Analytics Innovation Summit (2016)